

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
NASA SHARED SERVICES CENTER

RECOMMENDATION AND DETERMINATION TO SOLICIT FROM ONE SOURCE
OR BRAND NAME

PICK ONE

Sole Source:

☒ I recommend that NASA, NASA Shared Services Center, negotiate with Nikon AM Synergy Inc to provide an Ahead CP-1 Specimen NXG Build. The total estimated cost of this effort is [REDACTED], and the estimated period of performance is 8 weeks.

This recommendation is for the acquisition of supplies or services determined to be reasonably available from only one source/only one manufacture. Competition is impractical for the following reasons (please answer the questions below):

1. **What are the reasons for soliciting only the suggested source/suggested brand name?** Please provide the unique characteristics of the items or services along with the supporting rationale and the market research performed to substantiate your reason for the suggested source/brand name.

Marshall Space Flight Center (MSFC) requires manufacturing of Ahead CP-1 specimens on a Nikon SLM Solutions NXG additive manufacturing (AM) machine to evaluate properties of this alloy for potential use for producing manufacturing demonstrators of hardware. This machine is of interest for its large scale additive manufacturing capability with a 600mm x 600mm x 600 mm build volume that would be large enough to large complex components for NASA programs. Nikon AM Synergy has been identified as the only commercial vendor that is currently installed, qualified, and printing Ahead CP-1 on a Nikon SLM Solutions NXG machine.

2. **What is the impact to the government if another vendor/brand is used?** Include any dollar, schedule, or compatibility impacts if possible.

If another machine is chosen, the material produced will not be adequately represent the process to produce the targeted components and the performance of the material will not be directly comparable. This will leave a blind spot for the project in evaluating the viability of producing hardware using this large scale additive manufacturing system and identifying potential shortfalls of using an NXG platform. If a different vendor is chosen, additional costs to switch materials may be incurred that can be in excess [REDACTED] to perform as no other vendors have been identified as currently printing Ahead CP-1 on an NXG machine. This material switch would also add lead time which would impact NASA's schedule for FY26 milestones as a switch on such a large system would take multiple weeks to perform. NASA has engaged with Nikon AM Synergy (formerly Morf3D) under contract [REDACTED] to evaluate materials on a Nikon SLM Solutions NXG platform and continuity in material evaluation in identical machines and most experienced facilities is essential to improving NASA's technological understanding of large scale AM as this material will support efforts for STMD, SLS, and commercial space missions.

ADDITIONAL COMMENTS:

02/17/2026

Date

[REDACTED]